

Instructions: You will work on these exercises during the practical session in the sixth week of the course. We will cover exercises 1, 2a and 3. We encourage you to try to do them before the session.

Exercise 1: Valid Kernels

Which of the following are valid kernels? Justify your answer.

- (a) $k(\mathbf{x}, \mathbf{z}) = \|\mathbf{x} - \mathbf{z}\|_2^2$.
- (b) $k(\mathbf{x}, \mathbf{z}) = (2\mathbf{x}^\top \mathbf{z} + 1)^3$.

Exercise 2: The Kernel Trick

In the lectures, we have seen how the kernel trick extends linear models to nonlinear ones by replacing scalar products with a kernel function. In this exercise, you will apply this idea to different models.

- (a) The nearest-neighbour classifier assigns a new input vector $\mathbf{x}$ to the same class as that of the nearest training example $\mathbf{x}_n$. In the standard formulation, the distance is measured using the squared Euclidean distance

$$\|\mathbf{x} - \mathbf{x}_n\|^2.$$

Express this distance in terms of scalar products and derive the corresponding nearest-neighbour classifier for a general nonlinear kernel.

- (b) Consider the binary perceptron classifier with prediction rule

$$\hat{y} = \text{sign}(\mathbf{w}^\top \mathbf{x}),$$

and update rule

$$\mathbf{w} \leftarrow \mathbf{w} + y_n \mathbf{x}_n$$

whenever a training example $(\mathbf{x}_n, y_n)$ is misclassified.

First, show that after training the weight vector can be written as a linear combination of the vectors $y_n \mathbf{x}_n$. Then, derive the kernelized version of the prediction rule.

Note: assume we initialize $\mathbf{w} = \mathbf{0}$.

Exercise 3: Deriving the stationary and KKT conditions.

The purpose of this exercise is to show the reasoning behind the Lagrangian and KKT conditions from the method for solving inequality-constrained optimization problems introduced in the lectures.

Consider the constrained optimization problem:

$$\max_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) \quad \text{subject to} \quad g(\mathbf{x}) \geq 0.$$

We aim to find candidate points for a global maximum. The restriction $g(\mathbf{x}) \geq 0$ gives us the **primal feasibility** KKT condition. Restricting the domain to this region gives us two types of candidate points: those on the boundary $g(\mathbf{x}) = 0$ (Case I) and those in the region $g(\mathbf{x}) > 0$ (Case II).

- (a) (Case I) If there is a critical point $\mathbf{x}_0$ on the boundary, it means that an unconstrained maximum would occur either on the boundary or outside the feasible region, and the boundary restricts us from reaching that point.
- (i) In which direction does $\nabla_{\mathbf{x}}f(\mathbf{x}_0)$ point in this case?
 - (ii) What about $\nabla_{\mathbf{x}}g(\mathbf{x}_0)$?
 - (iii) How are the directions of both gradients related? Can you express $\nabla_{\mathbf{x}}f(\mathbf{x}_0)$ in terms of $\nabla_{\mathbf{x}}g(\mathbf{x}_0)$ using a scalar parameter μ ?
- (b) (Case II) If there is a critical point $\mathbf{x}_0$ inside the feasible region, then the inequality constraint does not restrict the location of the maximum of f .
- (i) What is the value of $\nabla_{\mathbf{x}}f(\mathbf{x}_0)$ in this case?
 - (ii) Can you express this condition using a similar equation as in (a.iii)?
- (c) Combine your results from (a.iii) and (b.ii) into a single condition that all candidate points must satisfy. Congratulations! You have derived the **stationarity condition** for the Lagrangian and the **dual feasibility** KKT condition.
- (d) Use your findings so far to derive the **complementary slackness** KKT condition.
- (e) How would we define the Lagrangian for the following optimization problems? *Hint: Think about whether $\nabla_{\mathbf{x}}f$ and $\nabla_{\mathbf{x}}g$ would be parallel or anti-parallel in each case.*
- (i) $\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x})$ subject to $g(\mathbf{x}) \geq 0$.
 - (ii) $\max_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x})$ subject to $g(\mathbf{x}) \leq 0$.
 - (iii) $\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x})$ subject to $g(\mathbf{x}) \leq 0$.

Exercise 4: Inequality-constrained optimization

Consider the following 1D optimization problem:

$$\min_{x \in \mathbb{R}} \frac{1}{2}(x-2)^2 \quad \text{subject to} \quad x \geq 4.$$

Of course, this is a very simple problem that can be solved just by evaluating the function in the critical point of the parabola ($x = 2$) and in the extreme of the domain ($x = 4$). However, we will take the long way around to further demystify the method of Lagrange multipliers.

- (a) Formulate the Lagrangian of the problem using a Lagrange multiplier $\mu \geq 0$ for the inequality constraint.
- (b) Derive the stationarity condition and express the primal variable x in terms of the dual variable μ .
- (c) Write down the complementary slackness condition.
- (d) Construct the Dual Lagrangian $\tilde{\mathcal{L}}(\mu) = \min_x \mathcal{L}(x, \mu)$ and formulate the dual problem.
- (e) Solve the dual problem to find μ^* and then compute the corresponding primal optimum x^* . Is the value located in the boundary or inside the constrained region?
- (f) Verify that the obtained solution meets the KKT conditions.